# Notes

## Session Overview: Relations and Functions

### Introduction

Hello everyone! Welcome to today's session. This session will be different from the previous two, which were more theoretical. Today, we will focus heavily on problem-solving.

### Session Objectives

We will target the following key areas:

- **Types of Relations:** We will begin by discussing the most prominent types of relations frequently asked in JEE, including:
  - Transitive relations
  - Equivalence relations
  - Symmetric relations
  - Reflexive relations
- **Functions:** After covering relations, we will move directly to functions, specifically focusing on domain-based problems.

### Expectations

I encourage everyone's active participation and cooperation. Stay focused, and I will ensure a smooth learning experience. We will be solving many questions, so practice hard!

## Representing Relations

### Methods of Representing a Relation

There are two primary ways to represent a relation:

1. **Listing all elements:** Explicitly write down every ordered pair that belongs to the relation.
2. **Defining a property/pattern:** Describe the rule or pattern that the elements in the ordered pairs follow. For example, if a relation follows the pattern $Y = 2X - 1$, then every ordered pair $(X, Y)$ in the relation satisfies this formula.

Both methods are standard and acceptable for representing a relation.

## Inverse of a Relation

### Defining the Inverse of a Relation ($R^{-1}$)

**How to find $R^{-1}$:**

If an ordered pair $(a, b)$ belongs to the relation $R$, then the ordered pair $(b, a)$ will belong to the inverse relation $R^{-1}$. In essence, the elements within each ordered pair are interchanged.

**Example:**

Let set A = {1, 2, 3} and set B = {5, 7}. Consider a relation $R$ from A to B, for instance, $R$ = $\{(1, 5), (2, 7)\}$.

To find $R^{-1}$:

- The ordered pair $(1, 5)$ in $R$ becomes $(5, 1)$ in $R^{-1}$.
- The ordered pair $(2, 7)$ in $R$ becomes $(7, 2)$ in $R^{-1}$.

So, $R^{-1}$ = $\{(5, 1), (7, 2)\}$.

**Domain and Range of Inverse Relation**

An important property of inverse relations is the interchange of domain and range:

- The **domain of $R$** becomes the **range of $R^{-1}$**.
- The **range of $R$** becomes the **domain of $R^{-1}$**.

## Types of Relations

**Introduction to Relations**

We will discuss various types of relations, which are crucial for understanding set theory and functions.

**Identity Relation**

An identity relation, denoted as $I_A$, is defined on a set A (from A to A). In an identity relation, every element of set A must be paired with itself, and *only* with itself.

- **Definition:** For a set $A$ = $\{1, 2, 3\}$, the identity relation $I_A$ would be $I_A$ = $\{(1, 1), (2, 2), (3, 3)\}$ .
- **Key Characteristics:**
  - Every element in set A must have a pair with itself.
  - No other pairs are allowed. If any extra pair is present, it is not an identity relation.
  - If any element from set A is missing its self-paired element, it is not an identity relation.

Example 2: If $A = \{1, 2, 3\}$ and a relation $R_2 = \{(1, 1), (2, 2)\}$, then $R_2$ is not an identity relation because the pair $(3, 3)$ is missing for the element $3 \in A$.

#### Reflexive Relation

A reflexive relation is also defined on a set A (from A to A). Similar to an identity relation, every element must be paired with itself. However, unlike an identity relation, a reflexive relation *allows* for other pairs to be present.

- **Definition:** A relation R on set A is said to be reflexive if every element of A is related with itself. That is, for every $a \in A$, $(a, a) \in R$.
- **Key Characteristics:**
  - Every element in set A must have a pair with itself.
  - Additional pairs are permitted. Their presence does not disqualify the relation from being reflexive.
- **Example 1:** If $A = \{1, 2, 3\}$ and a relation $R_1 = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3)\}$, then $R_1$ **is** a reflexive relation. The pairs $(1, 1), (2, 2), (3, 3)$ are present, and the extra pairs $(1, 2), (1, 3)$ are allowed.
- **Example 2:** If $A = \{1, 2, 3\}$ and a relation $R_2 = \{(1, 2), (1, 3)\}$, then $R_2$ is **not** a reflexive relation because the self-pairs $(1, 1), (2, 2), (3, 3)$ are missing.
- **Example 3:** If $A = \{1, 2, 3\}$ and a relation $R_3 = \{(1, 1), (2, 2), (3, 3)\}$, then $R_3$ **is** a reflexive relation. This is also an identity relation.

#### Relationship between Identity and Reflexive Relations

Every identity relation is an example of a reflexive relation. However, the converse is not true; not every reflexive relation is an identity relation.

## Recap: Reflexive Relation Example

#### Understanding Reflexive Relations with an Example

Let's revisit the concept of a reflexive relation with a specific example to ensure clarity.

Consider a relation $R_1$ defined on a set $A = \{1, 2, 3\}$. The relation $R_1$ is defined such that for any ordered pair $(X, Y)$, $Y \geqslant X$.

To determine the ordered pairs in $R_1$:

- If $X = 1$: $Y \geqslant 1$. The possible ordered pairs are $(1, 1), (1, 2), (1, 3)$.
- If $X = 2$: $Y \geqslant 2$. The possible ordered pairs are $(2, 2), (2, 3)$.
- If $X = 3$: $Y \geqslant 3$. The possible ordered pair is $(3, 3)$.

# Notes

A relation is reflexive if for every element $a$ in the set $A$, the ordered pair $(a, a)$ is present in the relation.

In our example, for set $A = \{1, 2, 3\}$, we need to check if $(1, 1)$, $(2, 2)$, and $(3, 3)$ are in $R_1$.

From the derived ordered pairs, we can see that:

- $(1, 1) \in R_1$
- $(2, 2) \in R_1$
- $(3, 3) \in R_1$

Since all elements of $A$ have their corresponding self-related pairs in $R_1$, the relation $R_1$ is indeed reflexive.

## Symmetric Relation Definition

### Definition of a Symmetric Relation

A relation $R$ on a set $A$ is said to be symmetric if, for all $a, b \in A$, whenever $(a, b) \in R$, then $(b, a)$ must also be in $R$.

In simpler terms:

- If an ordered pair $(a, b)$ exists in the relation, its reverse $(b, a)$ must also exist for the relation to be symmetric.
- This condition must hold true for *every* ordered pair in the relation.

### Example of Symmetric Relation Elements

Let's consider a set $A = \{1, 2, 3\}$. If a relation $R$ on $A$ is symmetric:

- If $(1, 2) \in R$, then $(2, 1)$ must also be in $R$.
- If $(2, 3) \in R$, then $(3, 2)$ must also be in $R$.
- If $(1, 3) \in R$, then $(3, 1)$ must also be in $R$.

If any of these conditions are not met, the relation is not symmetric. The relation between $A$ and $B$ must be the same as the relation between $B$ and $A$.

## Examples of Symmetric Relations

### Example 1: Parallel Lines

Consider a set of lines where the relation is "is parallel to".

line 2 ($L_2 \parallel L_1$).

- **Conclusion:** Yes, the relation "is parallel to" is a symmetric relation because if $(L_1, L_2)$ is in the relation, then $(L_2, L_1)$ is also in the relation.

#### Example 2: Perpendicular Lines

Consider a set of lines where the relation is "is perpendicular to".

- **Question:** Is the relation "L1 is perpendicular to L2" an example of a symmetric relation?
- **Analysis:** If line L1 is perpendicular to line L2 ($L_1 \perp L_2$), then it is also true that line L2 is perpendicular to line L1 ($L_2 \perp L_1$).
- **Conclusion:** Yes, the relation "is perpendicular to" is a symmetric relation because if $(L_1, L_2)$ is in the relation, then $(L_2, L_1)$ is also in the relation.

## Non-Symmetric Relation Example: Family Relations

#### Example: "Is the brother of" Relation

Consider a relation defined as "A is the brother of B".

- **Question:** Is the relation "A is the brother of B" an example of a symmetric relation?
- **Analysis:** If A is the brother of B, it does not necessarily mean that B is also the brother of A. B could be a sister.
  - For instance, if A is a male and B is a female, A is the brother of B. However, B is the sister of A, not the brother of A.
- **Conclusion:** No, the relation "is the brother of" is not a symmetric relation because if $(A, B)$ is in the relation, $(B, A)$ is not necessarily in the relation.

## Proving a Relation is Symmetric: Divisibility by 4

#### Problem Statement

Prove that the relation $a - b$ is divisible by 4, where $a, b$ belong to the set of integers, is a symmetric relation.

#### Understanding Symmetric Relations

A relation $R$ on a set $A$ is symmetric if, for all $a, b \in A$, whenever $(a, b) \in R$, then $(b, a) \in R$.

#### Given Relation

The given relation is defined as: $(a, b) \in R$ if and only if $a - b$ is divisible by 4.

#### Step-by-Step Proof

2. **Check for $(b, a) \in R$:** To prove the relation is symmetric, we need to show that if $a - b$ is divisible by 4, then $b - a$ is also divisible by 4. From the assumption, we have $a - b = 4m$. Now, consider $b - a$. We can write $b - a$ as $-(a - b)$. Substitute the expression for $a - b$: $b - a = -(4m)$ $b - a = -4m$

3. **Conclusion:** Since $m$ is an integer, $-m$ is also an integer. Therefore, $b - a = 4(-m)$ shows that $b - a$ is a multiple of 4. This implies that $b - a$ is divisible by 4. Thus, if $(a, b) \in R$, then $(b, a) \in R$.

#### Final Answer

Since we have shown that if $a - b$ is divisible by 4, then $b - a$ is also divisible by 4, the given relation is indeed a **symmetric relation**.

## Identifying Elements for Reflexive and Symmetric Relations

#### Problem Statement

Given a relation with elements $(1, 2)$, $(3, 4)$, and $(2, 3)$, identify the additional elements required to make the relation:

1. Reflexive
2. Symmetric

#### Understanding Reflexive Relations

A relation $R$ on a set $A$ is reflexive if for every element $a \in A$, $(a, a) \in R$. This means every element must be related to itself.

#### Understanding Symmetric Relations

A relation $R$ on a set $A$ is symmetric if, for all $a, b \in A$, whenever $(a, b) \in R$, then $(b, a) \in R$.

#### Given Elements

Let's assume the set $A$ contains elements $1, 2, 3, 4$ based on the given pairs. The existing pairs in the relation are: $(1, 2)$, $(3, 4)$, $(2, 3)$.

#### 1. Making the Relation Reflexive

For the relation to be reflexive on the set $\{1, 2, 3, 4\}$, the following pairs must be present:

- $(1, 1)$

- (4, 4)

From the given elements, $(1, 1)$ and $(2, 2)$ are already present (implied by the example, though not explicitly listed as given, the lecturer states "(1, 1) and (2, 2) are there"). Therefore, the additional elements required for reflexivity are:

- $(3, 3)$
- $(4, 4)$

#### 2. Making the Relation Symmetric

For the relation to be symmetric, for every pair $(a, b)$ in the relation, its reverse $(b, a)$ must also be present.

Let's check the given pairs:

- If $(1, 2)$ is present, then $(2, 1)$ must also be present.
- If $(3, 4)$ is present, then $(4, 3)$ must also be present.
- If $(2, 3)$ is present, then $(3, 2)$ must also be present.

Therefore, the additional elements required for symmetry are:

- $(2, 1)$
- $(4, 3)$
- $(3, 2)$

#### Summary of Required El

Did you find these notes helpful?

To make the relation both reflexive and symmetric, the following elements need to be added:

- For Reflexivity: $(3, 3)$, $(4, 4)$
- For Symmetry: $(2, 1)$, $(4, 3)$, $(3, 2)$

## Transitive Relation

#### Definition of Transitive Relation

A relation $R$ defined from set $A$ to $A$ is said to be transitive if, for any elements $x, y, z \in A$:

If $(x, y) \in R$ and $(y, z) \in R$, then it must follow that $(x, z) \in R$.

#### Key Logic for Verifying Transitivity

1. **If the pattern $(x, y)$ and $(y, z)$ exists in the relation:**

- If $(x, z)$ is *not* present, the relation is *not* transitive.

2. **If the pattern $(x, y)$ and $(y, z)$ does *not* exist in the relation:**
   - In this case, the condition for transitivity is vacuously true, and the relation is considered transitive by default.

**Examples of Transitive Relations**

Let's consider a set $A = \{1, 2, 3, 4, 5\}$.

**Example 1: Relation R1** Let $R1 = \{(1, 2), (2, 3), (1, 3)\}$.

- We observe the pattern $(x, y)$ and $(y, z)$: $(1, 2)$ and $(2, 3)$.
- Here, $x = 1$, $y = 2$, $z = 3$.
- According to the definition, $(x, z)$ which is $(1, 3)$ should be present.
- Since $(1, 3)$ is present in $R1$, this relation is **transitive**.

**Example 2: Relation R2** Let $R2 = \{(1, 2), (2, 3), (1, 3), (4, 5), (5, 1)\}$.

- First, check for $(1, 2)$ and $(2, 3)$. Here $x = 1$, $y = 2$, $z = 3$. $(1, 3)$ is present. This part is transitive.
- Next, check for $(4, 5)$ and $(5, 1)$. Here $x = 4$, $y = 5$, $z = 1$.
- According to the definition, $(x, z)$ which is $(4, 1)$ should be present.
- However, $(4, 1)$ is *missing* from $R2$.
- Since we found an instance where the condition is not met, $R2$ is **not transitive**.

**Example 3: Relation R3** Let $R3 = \{(1, 2), (3, 4)\}$.

- In this relation, there is no pair of the form $(x, y)$ and $(y, z)$ where the second element of the first pair is the same as the first element of the second pair.
- For example, we have $(1, 2)$, but no pair starts with $2$.
- We have $(3, 4)$, but no pair starts with $4$.
- Since the pattern $(x, y)$ and $(y, z)$ does not exist in $R3$, the relation is **transitive by default**.

## Proving Transitivity of a Relation: x is a factor of y

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## Quiz: Binary Relations on Set A

# Notes

## Solution: Verifying Symmetric Property

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## Solution: Verifying Transitive Property

### Transitive Property

A relation $R$ on a set $A$ is transitive if for every $(x, y) \in R$ and $(y, z) \in R$, it implies that $(x, z) \in R$.

1. **For $R_1$:**
   - Consider the pairs $(b, c) \in R_1$ and $(c, a) \in R_1$. For $R_1$ to be transitive, $(b, a)$ must be in $R_1$. However, $(b, a)$ is **missing** from $R_1$.
   - Therefore, $R_1$ is **not transitive**.
2. **For $R_2$:**
   - Consider the pairs $(a, b) \in R_2$ and $(b, a) \in R_2$. For $R_2$ to be transitive, $(a, a)$ must be in $R_2$. We can see $(a, a) \in R_2$.
   - Consider the pairs $(c, a) \in R_2$ and $(a, b) \in R_2$. For $R_2$ to be transitive, $(c, b)$ must be in $R_2$. However, $(c, b)$ is **missing** from $R_2$.
   - Therefore, $R_2$ is **not transitive**.

### Conclusion

- $R_1$ is not symmetric and not transitive.
- $R_2$ is symmetric but not transitive.

Comparing this with the given options:

- A: both $R_1$ and $R_2$ are not symmetric (Incorrect, $R_2$ is symmetric).
- B: $R_1$ is not symmetric but it is transitive (Incorrect, $R_1$ is not transitive).
- C: $R_1$ is symmetric but it is not transitive (Incorrect, $R_1$ is not symmetric).
- D: both $R_1$ and $R_2$ are transitive (Incorrect, neither is transitive).

There seems to be a discrepancy between the provided options and the derived properties. Let's re-evaluate the options based on our findings.

- $R_2$: Symmetric, Not Transitive

Let's re-check the options carefully:

- **Option B: $R_1$ is not symmetric but it is transitive.**
  - We found $R_1$ is not symmetric (Correct).
  - We found $R_1$ is not transitive (This part makes the option incorrect).

Let's re-examine the transitivity of $R_1$ for any other pairs. We found $(b, c) \in R_1$ and $(c, a) \in R_1$, but $(b, a) \notin R_1$. This confirms $R_1$ is not transitive.

Let's re-examine the transitivity of $R_2$. We found $(c, a) \in R_2$ and $(a, b) \in R_2$, but $(c, b) \notin R_2$. This confirms $R_2$ is not transitive.

Given the options, there might be an error in the question or options provided, as none perfectly match our derived properties. However, if we have to choose the *best* fit or if there's a subtle interpretation, let's re-evaluate.

The lecturer states: "R1 is not symmetric. R2 is symmetric." This matches our findings.

Then for transitivity:

- Lecturer on $R_2$: "But do I have BC? No. XYYZ pair is there but XZ pair is missing. That means it is not transitive, it is gone. So if I think about R2, it is symmetric but it is not transitive." This matches our finding for $R_2$.
- Lecturer on $R_1$: "In R1, you see BC is there and CA is there, but BA is missing." This matches our finding for $R_1$.

So, both $R_1$ and $R_2$ are not transitive. $R_1$ is not symmetric, and $R_2$ is symmetric.

Let's re-check the options with these confirmed properties:

- A: both $R_1$ and $R_2$ are not symmetric (False, $R_2$ is symmetric).
- B: $R_1$ is not symmetric but it is transitive (False, $R_1$ is not transitive).
- C: $R_1$ is symmetric but it is not transitive (False, $R_1$ is not symmetric).
- D: both $R_1$ and $R_2$ are transitive (False, neither is transitive).

There seems to be an issue with the provided options as none are fully correct based on the properties derived. However, the quiz results indicate a correct answer was chosen by over 60% of students. This suggests there might be a specific option intended as correct, possibly due to a slight misinterpretation or a common mistake. Without further clarification or a correct option provided, it's difficult to definitively select one from the given choices that aligns with the derived properties for both relations.

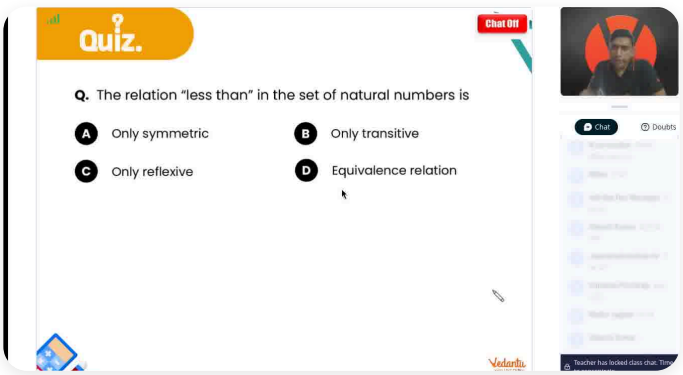
## Definition of Equivalence Relation

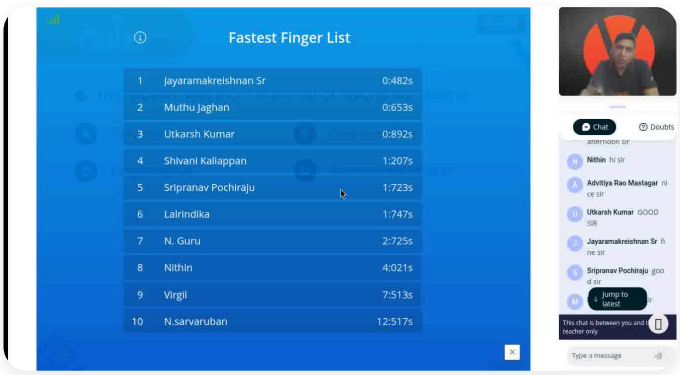
An equivalence relation is a type of binary relation that satisfies three key properties:

- **Reflexive:** Every element is related to itself.
- **Symmetric:** If element 'a' is related to element 'b', then element 'b' must also be related to element 'a'.
- **Transitive:** If element 'a' is related to element 'b', and element 'b' is related to element 'c', then element 'a' must also be related to element 'c'.

If a relation satisfies all three of these conditions (reflexive, symmetric, and transitive), it is called an **equivalence relation**.

## Quiz: Relation "less than" in Natural Numbers




### Quiz

Q. The relation "less than" in the set of natural numbers is

A. Only symmetric B. Only transitive C. Only reflexive D. Equivalence relation

### Solution Explanation

Let's analyze the given relation "less than" (<), defined on a set of rational numbers (or natural numbers, as stated in the quiz question).

1. **Symmetric Property:**
   - A relation $R$ is symmetric if for every $(x, y) \in R$, it implies $(y, x) \in R$.
   - In this case, if $x < y$, does it imply $y < x$? No.
   - For example, if we take $x = 1$ and $y = 2$, then $1 < 2$ is true. However, $2 < 1$ is false.
   - Therefore, the relation "less than" is **not symmetric**.
2. **Reflexive Property:**
   - A relation $R$ is reflexive if for every element $x$ in the set, $(x, x) \in R$.

- Therefore, the relation "less than" is **not reflexive**.

3. **Transitive Property:**
   - A relation $R$ is transitive if for every $(x, y) \in R$ and $(y, z) \in R$, it implies $(x, z) \in R$.
   - In this case, if $x < y$ and $y < z$, does it imply $x < z$? Yes.
   - This is a fundamental property of inequalities. If one number is less than a second, and the second is less than a third, then the first number must be less than the third.
   - Therefore, the relation "less than" is **transitive**.

Based on this analysis, the relation "less than" in the set of natural numbers is **only transitive**.

**Correct Option: B**

## Equivalence Relation for Similarity of Triangles

### Understanding Equivalence Relations

An equivalence relation is a type of binary relation that is reflexive, symmetric, and transitive.

### Checking for Reflexivity

- **Condition:** A relation R on a set A is reflexive if for every element $a \in A$, $(a, a) \in R$.
- **Application to Similarity:** For triangles, this means that any triangle T1 is similar to itself (T1 is similar to T1). This is always true.
- **Conclusion:** The similarity relation is reflexive.

### Checking for Symmetry

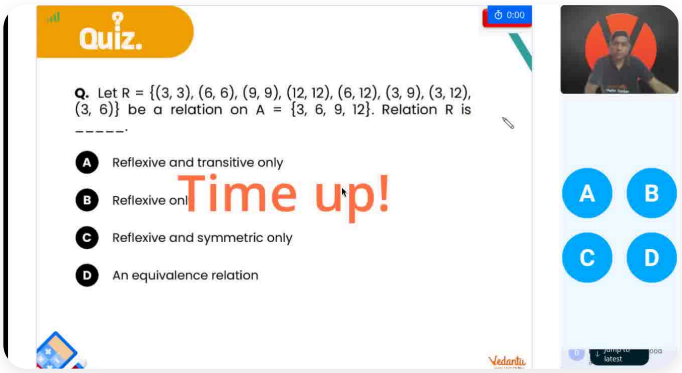
- **Condition:** A relation R on a set A is symmetric if for every $a, b \in A$, if $(a, b) \in R$, then $(b, a) \in R$.
- **Application to Similarity:** If triangle T1 is similar to triangle T2, then triangle T2 is also similar to triangle T1. This holds true.
- **Conclusion:** The similarity relation is symmetric.

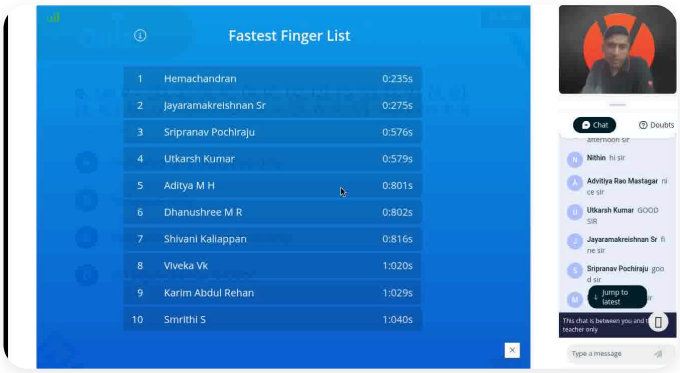
### Checking for Transitivity

- **Condition:** A relation R on a set A is transitive if for every $a, b, c \in A$, if $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$.
- **Application to Similarity:** If triangle T1 is similar to triangle T2, and triangle T2 is similar to triangle T3, then it logically follows that triangle T1 is similar to triangle T3.
- **Conclusion:** The similarity relation is transitive.

and transitivity), it is an equivalence relation.

## Quiz: Identifying Relation Types




### Question

Let $R$ = {(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)} be a relation on $A$ = {3, 6, 9, 12}. Relation R is:

A. Reflexive and transitive only B. Reflexive only C. Reflexive and symmetric only D. An equivalence relation

### Solution

To determine the type of relation R, we need to check for reflexivity, symmetry, and transitivity.

1. **Reflexivity:** A relation R on a set A is reflexive if for every element $a \in A$, $(a, a) \in R$. The set A is {3, 6, 9, 12}. We need to check if (3, 3), (6, 6), (9, 9), (12, 12) are all present in R. From the given relation R, we have:
   - $(3, 3) \in R$
   - $(6, 6) \in R$
   - $(9, 9) \in R$
   - $(12, 12) \in R$

   Since all elements $(a, a)$ for $a \in A$ are in R, the relation R is **reflexive**. This eliminates option B (Reflexive only) and D (An equivalence relation, which requires symmetry).

2. **Symmetry:** A relation R on a set A is symmetric if for every $(a, b) \in R$, it implies $(b, a) \in R$. Let's check the pairs in R:
   - Consider $(6, 12) \in R$. For symmetry, $(12, 6)$ must also be in R. However, $(12, 6) \notin R$.

   Therefore, the relation R is **not symmetric**. This eliminates option C (Reflexive and symmetric only) and D (An equivalence relation, which requires symmetry).

- Consider $(3, 6) \in R$ and $(6, 12) \in R$. For transitivity, $(3, 12)$ must be in R. We see that $(3, 12) \in R$.
- Consider $(3, 3) \in R$ and $(3, 6) \in R$. For transitivity, $(3, 6)$ must be in R. This is true.
- Consider $(3, 3) \in R$ and $(3, 9) \in R$. For transitivity, $(3, 9)$ must be in R. This is true.
- Consider $(3, 3) \in R$ and $(3, 12) \in R$. For transitivity, $(3, 12)$ must be in R. This is true.
- Consider $(6, 6) \in R$ and $(6, 12) \in R$. For transitivity, $(6, 12)$ must be in R. This is true.

After checking all possible combinations, the relation R is **transitive**.

Based on our analysis:

- R is Reflexive.
- R is not Symmetric.
- R is Transitive.

Therefore, the relation R is reflexive and transitive only.

#### Final Answer

The correct option is A.

## Introduction to Functions

#### Connecting Relations to Functions

A function is a special type of relation. A relation defined from set A to set B can be classified as a function if it satisfies two specific conditions:

1. **Every element of set A must have an image in set B.** This means that for every input from set A, there must be a corresponding output in set B. In simpler terms, no element in set A can be left unconnected.
2. **One value of X (from set A) cannot have two images in set B.** This implies that each element in set A must be mapped to exactly one element in set B. It's not possible for a single input to produce multiple different outputs.

#### Analogy for Understanding Functions

To better understand these conditions, consider the following analogy:

- Imagine elements of set A as soldiers from your country.
- Imagine elements of set B as soldiers from an enemy country.